

SA PrepBot — Design Summary

An AI-powered emergency preparedness assistant for San Antonio residents, answering questions on evacuation, shelters, supply kits, and severe-weather safety with cited, grounded answers drawn only from official City of San Antonio (COSA) sources.

1. System Architecture

The system is a retrieval-augmented generation (RAG) pipeline with a clear separation between offline data preparation and the online query path:

- **Data collection** — `scrape_cosa.py` crawls `sa.gov` emergency-preparedness pages with `crawl4ai` (a headless browser, needed because the site renders content client-side and blocks plain HTTP fetches), strips template boilerplate via marker-based trimming, and writes JSONL records.
- **Ingestion** — `ingestion.py` loads every `*.jsonl` file under `data/cosa_data/`, combining manually-curated and crawled batches into one document set.
- **Chunking & indexing** — `retrieval.py` splits documents into 1024-token chunks (50-token overlap) and embeds them into a local, persisted ChromaDB vector store.
- **Retrieval** — a query first pulls the top 20 chunks by vector similarity, then a cross-encoder reranker re-scores the (question, chunk) pairs and keeps the top 7.
- **Generation** — the top 7 chunks are passed to the LLM with a strict grounding prompt that requires inline citations for every factual sentence.
- **Citation formatting** — post-processes the LLM output: detects refusals, strips hallucinated URLs, renumbers citations to match the final source list, and returns structured answer + sources.
- **Serving** — a FastAPI backend exposes a single `/ask` endpoint; a Streamlit frontend provides the chat UI.

2. Technical Choices

- **LlamaIndex CitationQueryEngine** — chosen over a hand-rolled retrieval loop because it natively tracks which source chunks back which parts of the answer, which the citation-formatting step depends on.
- **ChromaDB (local, persisted)** — no external vector database infrastructure to run or pay for; the whole index lives in `src/storage/` and rebuilds from source JSONL in about a minute.
- **Two-stage retrieval (top-20 → rerank → top-7)** — pure vector similarity surfaces topically-related but sometimes weakly-relevant chunks; the `cross-encoder/ms-marco-MiniLM-L-12-v2` reranker scores the question against each candidate chunk jointly, which meaningfully improves precision on the final context window handed to the LLM.

- **gpt-5.4-mini at temperature 0** — OpenAI's current small/fast model, used for both answer generation and follow-up-question condensing. Temperature 0 favors deterministic, conservative answers over creative variation, which matters for a safety-information assistant.
- **FastAPI + Streamlit split** — the backend is a plain HTTP JSON API decoupled from the UI, so the frontend can be swapped (as it was, from React to Streamlit) without touching the pipeline.
- **crawl4ai for data collection** — sa.gov renders its content client-side and returns HTTP 403 to plain automated fetches; crawl4ai drives a real headless browser (Playwright) that renders the page like a normal visitor before content is extracted.

3. Prompt Templates

Answer-generation prompt (src/generator.py), enforcing strict grounding and a fixed refusal string:

You are the SA PrepBot Assistant, an emergency preparedness assistant for San Antonio residents.

Context information is provided below from official City of San Antonio (COSA) emergency preparedness sources.

```
-----
{context_str}
-----
```

Instructions:

1. Answer the query using ONLY the context provided.
2. Every sentence that uses information from the context MUST include an inline citation in the format [1], [2], etc. Do not write any factual sentence without a citation.
3. Do NOT include any URLs or hyperlinks in your answer. Reference sources using citation numbers only.
4. If the answer is not contained in the context, strictly say: 'I am sorry, but I cannot find supporting information in the indexed COSA documents.'

Query: {query_str}

Answer:

Follow-up condensing prompt — rewrites a conversational follow-up into a standalone question before retrieval, so multi-turn context doesn't get lost:

System: Given the conversation history and a follow-up question, rewrite the follow-up question as a fully self-contained standalone question. Do not answer the question - only rewrite it. If the follow-up is already standalone, return it unchanged.

User: Conversation:

```
{conversation history}
```

Follow-up question: {question}

Standalone question:

4. Architecture Adjustments from the Prior System

SA PrepBot's pipeline is a direct migration of an earlier academic-advisor RAG chatbot built for a different university/domain. The core mechanics (chunking, two-stage retrieval, citation formatting, multi-turn condensing) were kept as-is; the domain-facing layers were replaced:

- System prompt persona and refusal message rewritten for emergency preparedness / COSA, in place of the prior academic-catalog persona.
- ChromaDB collection renamed to reflect the new domain, so the two projects' indexes never collide if run side-by-side.
- Ingestion generalized from a single fixed JSONL file to loading every JSONL file in a data directory, since COSA source data now arrives as multiple batches (a small hand-curated set plus ongoing crawler output) rather than one static catalog export.
- A new data-collection stage (`scrape_cosa.py`) was added, since the prior project's source data was provided pre-scraped by a university department, whereas COSA data had to be collected first. Two content-boilerplate templates were identified across `sa.gov`'s page layouts and handled with per-template header/footer marker trimming, after a density-based content filter (`crawl4ai`'s `PruningContentFilter`) proved unreliable on this site's nested markup.
- Frontend replaced (React → Streamlit) for faster iteration without a Node/JS toolchain, restyled to keep the prior UI's visual language (translucent chat panel, colored message bubbles, inline citation links, hover-preview sources, typing indicator) using solid colors in place of image assets that weren't carried over.
- LLM updated to OpenAI's current small model (`gpt-5.4-mini`) in place of the model used at the time of the original build.

5. Known Limitations

- Data coverage is not exhaustive: `311.sanantonio.gov`'s knowledge-base articles redirect to a broken portal endpoint and are not currently indexed; PDF resources (e.g. the SAOEM Basic Plan, Winter Weather Preparedness Guide) are not yet ingested.
- The vector index is rebuilt from scratch on each run of `retrieval.py` rather than incrementally updated, which is fine at the current data volume but would need revisiting at scale.